

```

i = 1
sum = 0
x = 10
y = 5

L1:
if i > 100 goto L2

t1 = x * y
t2 = i * 4
t3 = t2 + a
M[t3] = t1
t4 = x * y
sum = sum + t4

t5 = i + 1
i = t5
goto L1

L2:
result = sum
print (result)

```

① 划分基本块

B₁:

```

i = 1
sum = 0
x = 10
y = 5

```

B₂:

```

L1:
if i > 100 goto L2

```

B₃:

```

t1 = x * y
t2 = i * 4
t3 = t2 + a
M[t3] = t1
t4 = x * y
sum = sum + t4
t5 = i + 1
i = t5
goto L1

```

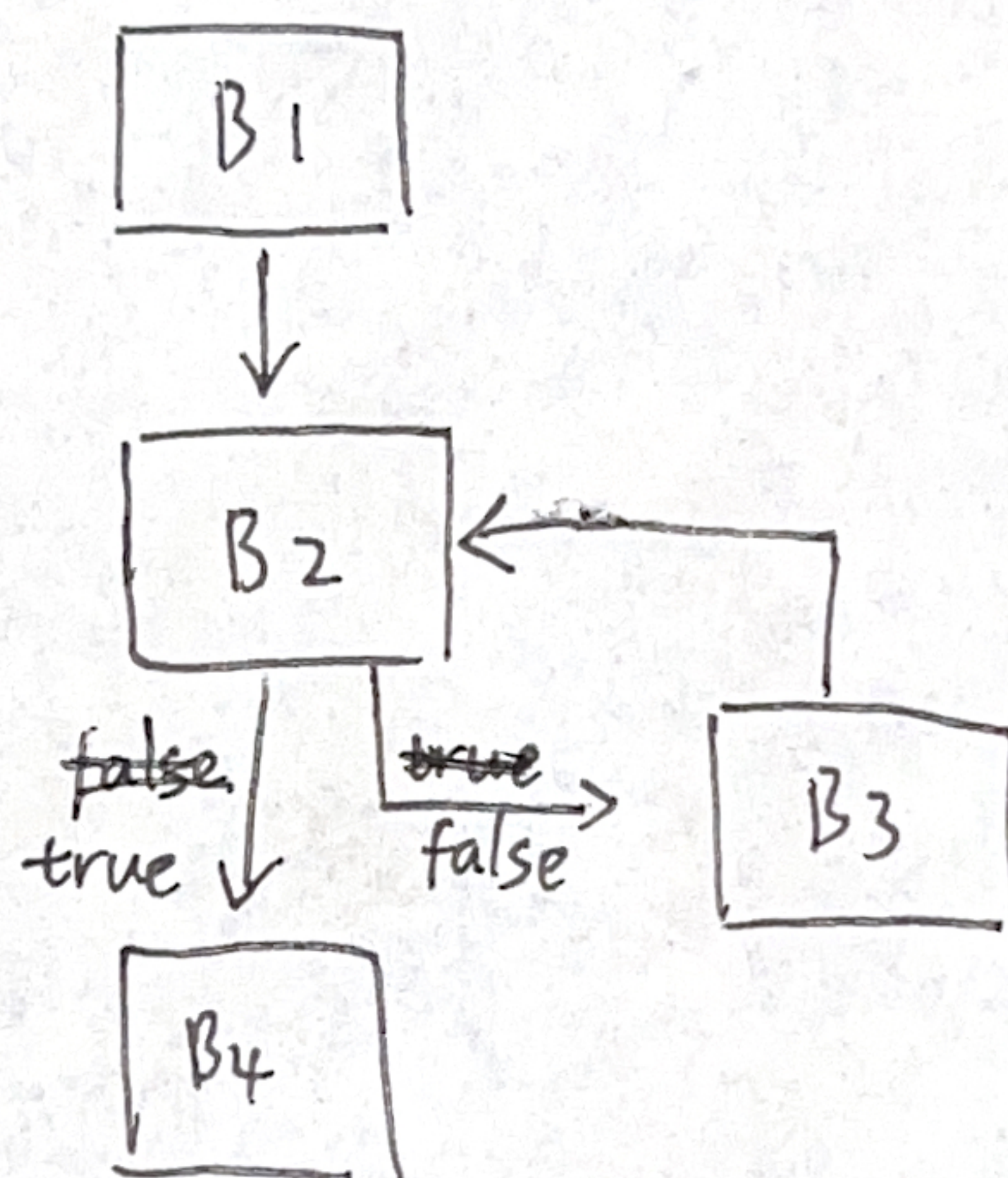
B₄:

```

L2:
result = sum
print (result)

```

② 构造流图



③ SSA代码构造

B1: $i_0 = 1$

$sum_0 = 0$

$x_0 = 10$

$y_0 = 5$

B2: $i_1 = \phi(i_0, i_2)$

$sum_1 = \phi(sum_0, sum_2)$

if $i_1 > 100$ goto ~~B4~~ B4

B3: $t1_0 = x_0 * y_0$

$t2_0 = i_1 * 4$

$t3_0 = t2_0 + a$

$M[t3_0] = t1_0$

$t4_0 = x_0 * y_0$

$sum_2 = sum_1 + t4_0$

$t5_0 = i_1 + 1$
 $i_2 = \cancel{t1_0} + t5_0$

goto ~~B2~~ B2

B4: $result_0 = sum_1$
 $print(result_0)$

④ 代码优化:

<1> 公共子表达式删除

$t1_0 = x_0 * y_0$

$t4_0 = x_0 * y_0$

$sum_2 = sum_1 + t4_0$

这里删去 $t4_0$ 定义, 将 $t4_0$ 换为 $t1_0$

<2> 循环不变量外提

$x_0 * y_0$ 为不变量

将 $t1_0$ 提到 pre-head 中, 即 B1

<3> 复制传播

$result_0$ 复制 sum_1

$t5_0 = i_1 + 1$

$i_2 = \cancel{t5_0} + 1$

可改为 $print(sum_1); \Rightarrow i_2 = i_1 + 1$

<4> ~~常量传播~~

~~$i_1 = \phi(1, i_2)$~~

~~$sum_1 = \phi(0, sum_2)$~~

~~x_0, y_0 换为 10 和 5~~

~~此时可删除 i_0, sum_0, x_0, y_0 定义~~

~~此时 $t1_0 = 50$, 也可替换常量~~

<5> 删除归纳变量

$t2_0$ 和 i_2 为“同步”的归纳变量

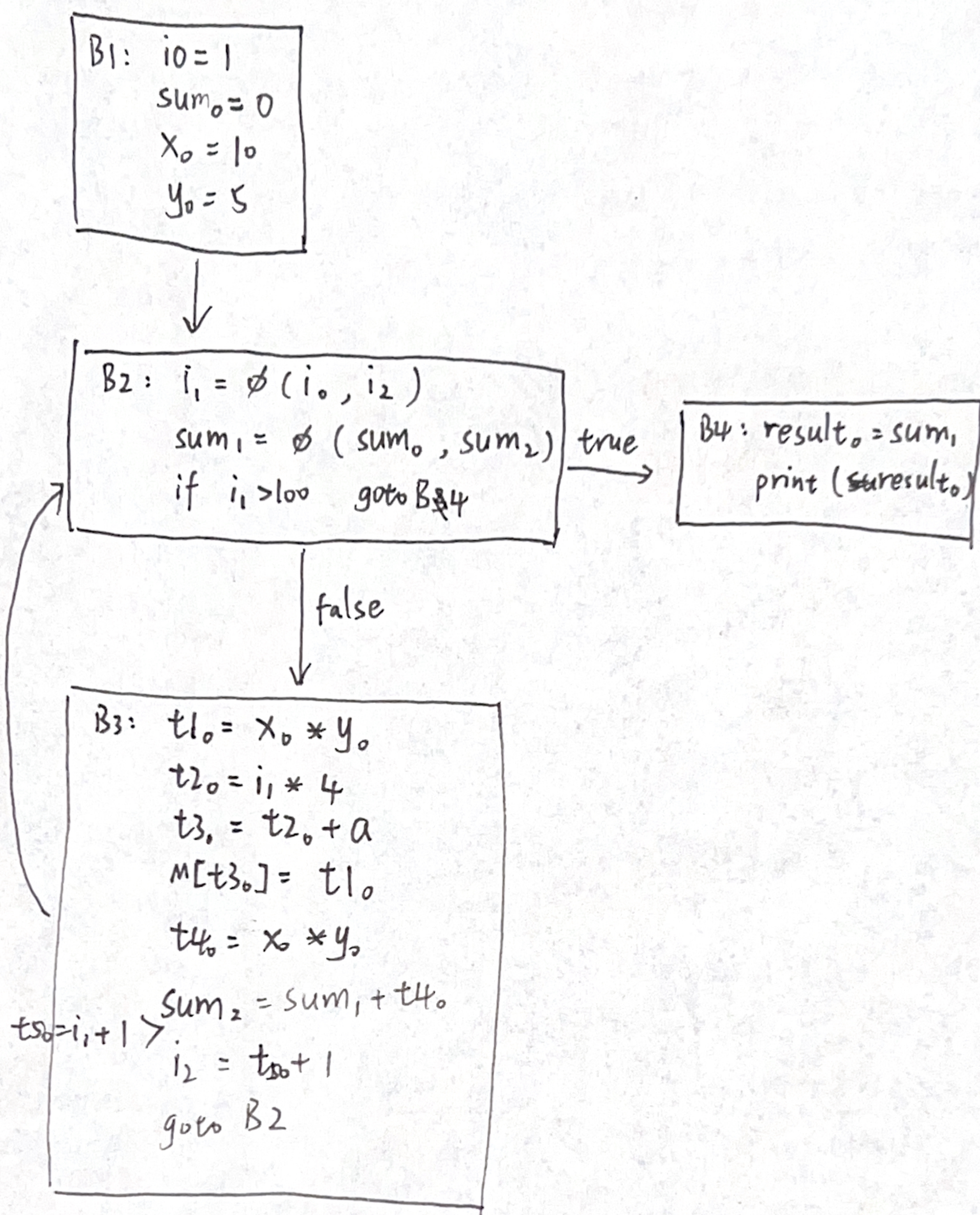
这里删去 i_2 和 i_1 , 并修改 $i_1 > 100$ 条件

<6> 强度削弱

将 $t2_0 = i_1 * 4$ 削弱为 $t2_0 = \cancel{t2_0}^{t2_1} + 4$

但要对 $t2_1$ 做初始值定义, ~~$t2_1 = 4$~~

⑤ SSA形式的流程图



⑥ 优化后流程图 SSA

